

MB 401: Mathematics for Business Management

Course Code	MB 401
Course Name	Mathematics for Business Management
L-T-P-C	3-0-0-3
Prerequisites	None
Intended for	Integrated MBA
Mutual Exclusion	NA
Approval	59th BoA (Apr 2025), in the Integrated MBA (IMBA) curriculum; 65th BoA (Aug 2026)

Preamble

Data-driven decision-making is increasingly central to effective management and business leadership. As organisations generate and leverage growing volumes of data, managers and business leaders need more than technical familiarity with analytics, i.e., they need the ability to understand the quantitative logic behind analytical insights, evaluate evidence critically, and translate analysis into sound business decisions. This course develops the mathematical and statistical foundations required to engage confidently with business analytics, machine learning, and artificial intelligence.

The course begins by establishing the fundamental concepts of variables, relationships, and data-generating processes, providing students with a structured way to understand and frame business problems. It introduces dimensional thinking as a managerial and analytical framework for representing business situations across multiple, interconnected dimensions and for understanding how different factors influence business outcomes.

A central focus of the course is the integration of business reasoning with mathematical thinking. Students learn to move from a real-world managerial problem to its analytical representation, identify the relevant variables and relationships, and use appropriate quantitative approaches to evaluate alternative courses of action. This integration enables students to appreciate mathematics not as an abstract discipline, but as a practical language for structuring business problems and improving managerial judgment.

The course also develops foundational knowledge of statistics, probability, and uncertainty, enabling students to interpret data, assess the strength of evidence, understand variability and risk, and make decisions when information is incomplete or outcomes are uncertain. Through a business-oriented perspective, students learn to distinguish between data, evidence, and actionable insight, and to recognise the implications of quantitative analysis for managerial decision-making.

By the end of the course, students will be able to view business challenges from multiple interacting dimensions, apply quantitative reasoning to managerial problems, and critically engage with analytical outputs. The course provides the conceptual foundation necessary for subsequent learning in business analytics, data science, machine learning, and artificial intelligence, while strengthening the broader managerial capability to make structured, evidence-based decisions in complex and uncertain business environments.

Objectives of the Course

On completion of this course, the student should be able to:

- Develop a strong foundation in the mathematical and statistical concepts underlying business analytics and data-driven decision-making.
- Apply dimensional thinking to represent and analyze complex business problems in multidimensional analytical spaces.
- Use concepts from linear algebra and calculus to model relationships and understand analytical and machine learning methods.
- Apply optimization techniques to formulate and solve managerial decision problems under constraints.
- Interpret statistical and probabilistic information to evaluate uncertainty, draw meaningful inferences, and support evidence-based business decisions.

Course Modules

- **Introduction:** Describing Variables and Relationships: Types of variables and their distribution, Theoretical distributions; what is a relationship? Conditional distributions, conditional means, line fitting, data generating process and identification. (4 hours)
- **Dimensional Thinking and Business Analysis:** Dimensional Thinking as a managerial mindset- concept of dimensions, understanding multi-dimensional space, multidimensional space vs. analysis space, transforming a business situation to a dimensional space, multidimensional analysis, dimensions in Euclidean and non-Euclidean space. (6 hours)
- **Linear Algebra in Dimensional Space:** Vectors, matrices and their operations in dimensional space; understanding linear Independence and basis vector concepts in dimensional space; rank and dimensionality, concepts of Tensors; dimensionality reduction. Concepts of linearity and nonlinearity, linear Mappings, Vector and matrix norms; Eigenvectors and eigenvalues, singular value decomposition (SVD). (10 hours)
- **Calculus in higher-dimensions:** Functions, relationships and dimensions; Linear and nonlinear functions, univariate and multivariate functions, basic functions, squashing functions and activation functions; composite functions. Derivatives and gradients, derivative of basic functions and activation functions, Chain rule; Partial derivatives, derivatives of multivariate functions; Optimization concepts; Integrals, substitution rule, areas between curves. (6 hours)
- **Optimization in Multiple Dimensions:** Concepts of single and multiple attributes, objectives and criteria; Concepts of constraints and constraint satisfaction problems; Maximum and minimum of univariate and multivariate functions, Saddle Points, local and global optimum, concepts of linear and nonlinear optimization, constrained and unconstrained optimization; search space, feasible and infeasible solution space, single agent and multi-agent search problems, search domain exploration and exploitation; Least squares method, Concept of gradient, gradient of vector valued functions, gradient of matrices; loss functions, gradient descent method. (8 hours)
- **Statistics, Probability and Uncertainty:** Statistics as a Language; Probability and common mistakes, conditional probability and Bayes' thinking in real life; Averages, Percentages, Rates, and Risk; Data Collection, Sampling, and Bias; Practical Framework for Evaluating Statistical Claims. (8 Hours)

Laboratory Modules

None

Text Books

1. Mike Cohen (2022), *Practical Linear Algebra for Data Science: From Core Concepts to Applications Using Python*, Shroff Publishers & Distributors Pvt. Ltd.
2. Marc Peter Deisenroth, A. Aldo Faisal, and Cheng Soon Ong (2020), *Mathematics for Machine Learning*, Cambridge University Press (<https://mml-book.com>).

References

1. Gilbert Strang (2025), *Linear Algebra and Learning from Data*, NAV Prints.
2. David C. Lay, Steven R. Lay, Judi J. McDonald (2016), *Linear Algebra and Its Applications*, Pearson Education.
3. James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021), *An Introduction to Statistical Learning with Applications in R*, Springer.
4. Thomas Nield (2022), *Essential Math for Data Science: Take Control of Your Data with Fundamental Linear Algebra, Probability, and Statistics*, Shroff Publishers & Distributors Pvt. Ltd.
5. Charu C. Aggarwal (2020), *Linear Algebra and Optimization for Machine Learning*, Springer Nature Switzerland AG.
6. *MS Excel 2019* (2020), *Data Analysis and Business Modeling*, Wayne Winston, Microsoft Press (PHI).